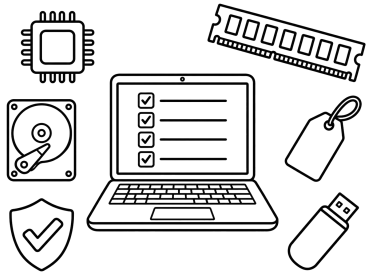
TEACHER GUIDE

ANSWERS AND MARKS

GRADE 9 · INFORMATION AND COMMUNICATION TECHNOLOGY

UNIT 1
Preparation of Computer Specifications



! FOR THE TEACHER ONLY

This booklet has the answers to every exercise in the Unit 1 Student Workbook, in the same order. The exercise letters match the workbook.

i WHAT IS IN THIS GUIDE

- **Marks and pacing** - every exercise and its marks
- **Class marks record** - one page to copy for each class
- **Practicals at a glance** - what to check, safety, a quiz, a practical test
- **Answers** - chapters 1 to 4, in the same order as the workbook
- **Unit 1 Test** - the marking scheme and the common mistakes
- **Extra** - textbook links, homework ideas and teaching tips

★ BEFORE THE FIRST LESSON

- ☐ Computer advertisements, price lists and a warranty card are collected.
- ☐ You can open This PC - Properties and File Explorer on the lab computers.
- ☐ A pen drive (checked for viruses) is ready for Practical 8.

"Bring real computer advertisements to the class. Students learn specifications best from real life."

- THE KING OF ICT

TEACHER GUIDE

Marks and Pacing

You decide how much class time each part needs

Chapter	Exercise	Marks
Chapter 1	A Match the users	5
	B Non-portable or mobile?	6
	C Input, output or storage?	8
	D Name the computers	6
	E True or false	6
Chapter 2	A Fill in the blanks	8
	B MHz and GHz	6
	C Which processor is faster?	5
	D Complete the processor table	9
	E True or false	6
Chapter 3	A MB and GB	6
	B Match the examples	6
	C Complete Figure 1.17	4
	D VGA, DVI or HDMI?	5
	E Which sound port?	4
	F Free, licensed or trial?	6
	G True or false	6
Chapter 4	A Which warranty is it?	6
	B Technical or non-technical?	8
	C Name the ports	6
	D Complete Figure 1.19	8
	E Compare two computers	6
End	Unit 1 Test	40
TOTAL		**176**

! SUGGESTED SHAPE OF ONE CHAPTER

READ	Teach the chapter pages. Students read with you and underline the key words.
DO	The practical tasks on real computers, in pairs.
WRITE	The written exercises. Walk around and help.
MARK	Mark the work together. Fill the tick boxes.

★ TEACHING TIPS

- Bring real computer advertisements and price lists to every lesson. They are the best teaching aids for this unit.
- Measure a lab monitor diagonally (corner to corner) with a tape that shows inches.

TEACHER GUIDE

Class Marks Record

Copy this page for each class

No	Student name	Ch 1 (31)	Ch 2 (34)	Ch 3 (37)	Ch 4 (34)	Test (40)	Total (176)
1							
2							
3							
4							
5							
6							
7							
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11							
12							
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25							
26							
27							
28							
29							

TEACHER GUIDE

Practicals at a Glance

The workbook activities 1.1 - 1.11 of the textbook

Practical	Students do	Check that ...
1 (Act. 1.1)	Find the types of computers at school and in advertisements	each computer is named (Figure 1.2) and called non-portable or for mobile use.
2 (Act. 1.2)	List the peripheral devices of a lab computer	every device is marked input, output or storage; the touch screen is both.
3 (Act. 1.3)	Find the processor and its speed (This PC - Properties)	the speed is written in GHz and in MHz; the type comes from Table 1.4.
4 (Act. 1.4)	Find the size and the free space of the hard disk	Local Disk (C:) and any other drives are written with their sizes.
5 (Act. 1.5)	Find the installed RAM	the size is written in GB and in MB (1 GB = 1024 MB).
6 (Act. 1.6, 1.7)	Compare two advertisements; compare one price at three places	the reasons for a higher price come from page 17; the warranty is also checked.
7 (Act. 1.8)	Find the after-sales services of two vendors	technical advice, technical assistance and the contact information are found.
8 (Act. 1.9)	Count the USB ports; connect a pen drive	the pen drive is opened and removed safely; devices are named from the list.
9 (Act. 1.10)	Sort an advertisement into technical and non-technical	the tree looks like Figure 1.19, with an example value for each specification.
10 (Act. 1.11)	Prepare a specification for a chosen user	each choice comes from the table of page 20 and suits the user requirements.

SAFETY AND MATERIALS

- Collect computer advertisements, price lists and warranty cards before the unit starts.
- Practicals 3 - 5: students only read the System and This PC windows. They change nothing.
- Practical 8: check the pen drive for viruses first, and show how to eject it safely.
- Old parts to show (a processor, a RAM card, a hard disk) are held by their edges only.

Textbook links

Workbook	Textbook pages	Activities
Chapter 1 - Users, computers and peripheral devices	1 - 6	1.1, 1.2
Chapter 2 - Specifications and the processor	6 - 10	1.3
Chapter 3 - Hard disk, monitor, memory and software	10 - 15	1.4, 1.5
Chapter 4 - Buying the right computer	16 - 20	1.6 - 1.11
Unit 1 in One Page	20 - 21 (summary)	-

TIP: Let pairs swap books and check each other's tick tables before you mark them.

? QUICK ORAL QUIZ - START OR END A LESSON WITH IT

Ask the class ...	Answer
1. Who develops and maintains websites?	**A Web Developer**
2. Name two computers for mobile use.	**Laptop, notebook, tablet, smart phone**
3. Is a touch screen input or output?	**Both**
4. Which part is the "brain" of the computer?	**The processor (CPU)**
5. How many MHz make 1 GHz?	**1000**
6. How many processors has a dual core CPU?	**Two**
7. How is the size of a monitor measured?	**Diagonally, in inches**
8. How many MB make 1 GB?	**1024**
9. Which port sends sound and video?	**HDMI**
10. Which warranty brings a technician to you?	**On-site warranty**

> PRACTICAL TEST - 30 MINUTES, 20 MARKS (OPTIONAL)

In the lab, one student at a time. Use a computer that is switched on, and one advertisement.

Step	The student must ...	Marks
1	Name the input, output and storage devices of the computer on the table.	4
2	Find the processor name and speed, and write the speed in MHz too.	4
3	Find the installed RAM and the size of Local Disk (C:).	4
4	Point to a USB port, the network (RJ45) port and the sound ports.	3
5	Sort the specifications of an advertisement into technical and non-technical.	5
	Total	**20**

★ TEACHING TIP

- Count the USB ports of a lab computer together. Ask which devices in the lab use them.

ANSWERS

Chapter 1

Users, computers and peripheral devices

A Match the users

5 marks

1. Programmer → **C**	**2.** Network Administrator → **E**
3. System Analyst → **A**	**4.** Computer Application Assistant → **D**
5. Web Developer → **B**	

B Non-portable or mobile?

6 marks

The computer	Non-portable	Mobile use
Server computer	☑	☐
Laptop computer	☐	☑
All-in-one computer	☑	☐
Smart phone	☐	☑
Workstation	☑	☐
Tablet computer	☐	☑

C Input, output or storage?

8 marks

The device	Input	Output	Storage
Keyboard	☑	☐	☐
Monitor	☐	☑	☐
Flash drive	☐	☐	☑
Microphone	☑	☐	☐
Printer	☐	☑	☐
Hard disk drive	☐	☐	☑
Speaker	☐	☑	☐
Touch screen	☑	☑	☐

1 mark each. The touch screen needs both ticks (input and output) for its mark.

D Name the computers

6 marks

1. Laptop computer	**2.** Server computer	**3.** Smart phone
4. All-in-one computer	**5.** Tablet computer	**6.** Desktop computer

1 mark each. Accept "laptop", "server", "tablet" without the word "computer". Accept "notebook" for 1.

E True or false

6 marks

1. **T - A Web Developer develops and maintains websites.**
2. **T - Users can be classified into system users and end users.**
3. **F - Server computers are computers for mobile use.**
4. **T - Computers for mobile use can be used in buses, trains and aeroplanes.**
5. **F - A printer is an input device.**
6. **T - A touch screen is both an input device and an output device.**

ANSWERS

Chapter 2

Specifications and the processor

A Fill in the blanks

8 marks

1. quality	**2.** specifications	**3.** brain	**4.** data
5. instructions	**6.** Hertz	**7.** AMD	**8.** capacity

B MHz and GHz

6 marks

1. 2 GHz = **2000 MHz**	**2.** 2.8 GHz = **2800 MHz**	**3.** 3.4 GHz = **3400 MHz**
4. 3000 MHz = **3 GHz**	**5.** 3200 MHz = **3.2 GHz**	**6.** 500 MHz = **0.5 GHz**

GHz to MHz: multiply by 1000. MHz to GHz: divide by 1000.

C Which processor is faster?

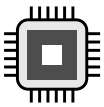
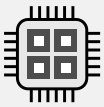
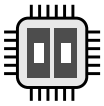
5 marks

1. 2.8 GHz or 3.4 GHz: **3.4 GHz**	**2.** 3.2 GHz or 3000 MHz: **3.2 GHz**
3. 2800 MHz or 3.0 GHz: **3.0 GHz**	**4.** 3400 MHz or 3.2 GHz: **3400 MHz**
5. 1 GHz or 900 MHz: **1 GHz**	

Change both speeds to the same unit first: 3.2 GHz = 3200 MHz, 3.0 GHz = 3000 MHz, 1 GHz = 1000 MHz.

D Complete the processor table

9 marks

The chip	No. of CPUs	Type	Name (Intel)
	**1**	**Single Core**	**Pentium I, II, III or IV**
	**4**	**Quad Core**	**Core i3, i5, i7 or i9**
	**2**	**Dual Core**	**Dual Core or Core 2 Duo**

1 mark for each cell. For the name accept any one of the names in that row of Table 1.4.

TEACHING TIP

- Let students name the job of each user in Table 1.1. Ask who maintains the school network.

E True or false

6 marks

1. **T - Specifications help us to know the quality of an item.**
2. **F - A maths exercise book and an art drawing book have the same specifications.**
3. **T - A machine working at a higher speed does more work in a unit time.**
4. **F - 1 GHz is equal to 100 MHz.**
5. **T - A dual core processor has two processors in it.**
6. **F - Core i3, i5, i7 and i9 are single core processors.**

ANSWERS

Chapter 3

Hard disk, monitor, memory and software

A MB and GB

6 marks

1. 1 GB = **1024 MB**	**2.** 2 GB = **2048 MB**	**3.** 4 GB = **4096 MB**
4. 8 GB = **8192 MB**	**5.** 2048 MB = **2 GB**	**6.** 512 MB = **0.5 GB**

GB to MB: multiply by 1024. MB to GB: divide by 1024 (512 MB is half a GB).

B Match the examples

6 marks

1. The brain of a human → **C**	**2.** A blender working at a higher speed → **F**
3. A 160 page exercise book → **E**	**4.** A wall clock that is easy to see → **A**
5. A rack for school bags in a library → **D**	**6.** The length, width and pages of a book → **B**

C Complete Figure 1.17

4 marks

1. User	**2.** Application software	**3.** Operating system	**4.** Hardware

1 mark each, from the top: the user, then application software, the operating system and the hardware.

TEACHING TIPS

- Start with the textbook story: one student needs a computer for documents and accounts, the other a mobile one for graphic designs. Ask the class what each should buy.
- Sort device cards into input, output and storage on the board. Keep the touch screen for the end.

D VGA, DVI or HDMI?

5 marks

The port ...	VGA	DVI	HDMI
It sends both sound and video signals.	☐	☐	☑
It sends only images, not sound.	☑	☐	☐
Digital Visual Interface	☐	☑	☐
High Definition Multimedia Interface	☐	☐	☑
Video Graphic Array (or Adapter)	☑	☐	☐

E Which sound port?

4 marks

1. Speakers: **Light green**	**2.** A microphone: **Light pink**
3. Headphones: **Light green**	**4.** Sound from an external device (line in): **Light blue**

F Free, licensed or trial?

6 marks

The operating system	Free and open source	Licensed	Trial version
Linux	☑	☐	☐
Windows 10 bought from a shop	☐	☑	☐
Ubuntu	☑	☐	☐
Windows 8	☐	☑	☐
It can be used only for a limited period.	☐	☐	☑
It can be obtained free of charge.	☑	☐	☐

G True or false

6 marks

1. **T - The hard disk stores data and software permanently.**
2. **F - A 1 TB hard disk can store more data than a 4 TB hard disk.**
3. **F - The size of a monitor is measured from its left side to its right side.**
4. **F - CRT monitors use less electricity than LCD and LED monitors.**
5. **T - The main memory stores data and instructions temporarily.**
6. **T - A separate VGA card has its own video memory and processor.**

★ TEACHING TIPS

- Show a real exercise book and a drawing book. Let students measure them: that is a specification.
- Show the RJ45 cable of a lab computer. Ask what would happen if it were unplugged.

ANSWERS

Chapter 4

Buying the right computer

A Which warranty is it?

6 marks

The warranty ...	Manufacturer	Extended	On-site
It is given by the manufacturing company.	☑	☐	☐
Given in addition to the standard warranty.	☐	☑	☐
Technicians come to your office to repair it.	☐	☐	☑
The customer pays an additional cost for it.	☐	☑	☐
A faulty device is replaced with a new one.	☑	☐	☐
Faulty parts are replaced at your workplace.	☐	☐	☑

B Technical or non-technical?

8 marks

The specification	Technical	Non-technical
Processor speed	☑	☐
Warranty	☐	☑
Memory capacity	☑	☐
Company reputation	☐	☑
Screen size	☑	☐
After sales service	☐	☑
Hard disk capacity	☑	☐
Computer price	☐	☑

C Name the ports

6 marks

1. Network port (RJ45)	**2.** HDMI port	**3.** USB port
4. Sound ports (audio jacks)	**5.** VGA port	**6.** DVI port

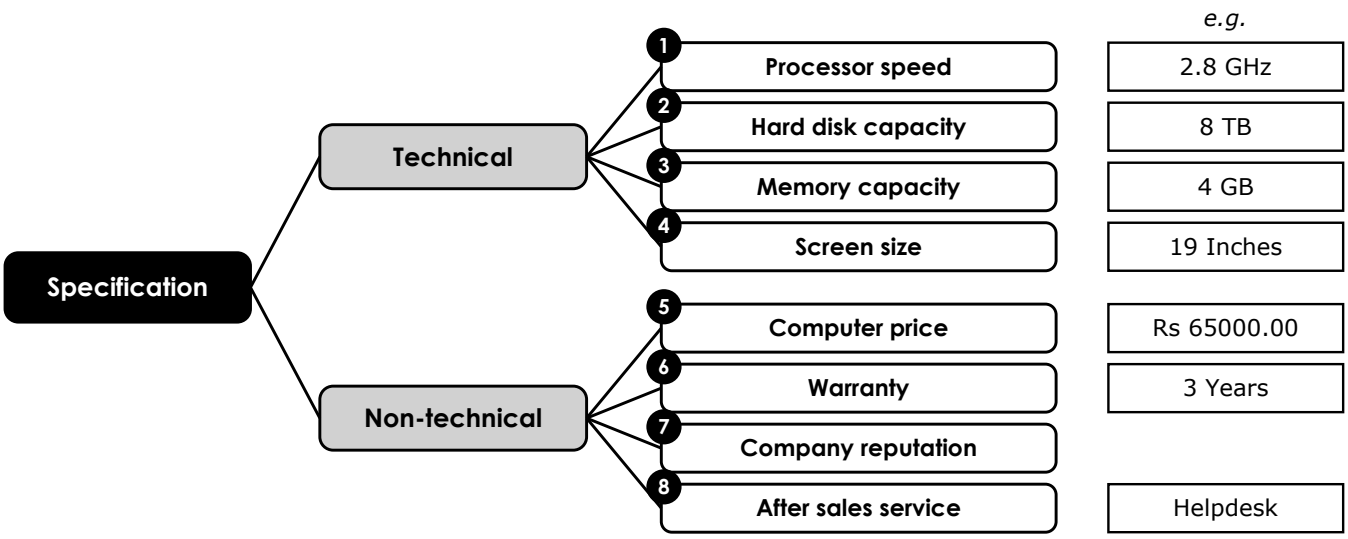
1 mark each. Accept "Ethernet port" or "LAN port" for 1 and "audio ports" for 4.

TEACHING TIPS

- Use the blender example with a stopwatch: the faster machine does more work in the same time.
- Write 1000 MHz = 1 GHz and 1024 MB = 1 GB on the board for the whole unit. Students mix them up.
- Keep a poster of Figure 1.19 and the table of page 20 on the wall during the unit.

D Complete Figure 1.19

8 marks



1 mark each. Boxes 1 - 4 are technical and 5 - 8 non-technical; the examples show which name goes in which box.

E Compare two computers

6 marks

1. **Which computer has the faster processor? (1 mark)**
   Computer B (3.4 GHz).
2. **Which computer can store more data? (1 mark)**
   Computer B (a 2 TB hard disk).
3. **Which computer is better for playing computer games? Why? (1 mark)**
   Computer B: it has a separate VGA card (with its own video memory and processor).
4. **Which computer has a free and open source operating system? (1 mark)**
   Computer A (Linux).
5. **Write two reasons why Computer B costs more. (2 marks)**
   Any two: a high speed processor, a high capacity hard disk, more memory, a large monitor, a licensed operating system, a longer (on-site) warranty. 1 mark each.

TEACHING TIPS

- Show an old processor, a RAM card and a hard disk if the school has them. Hold them by the edges.
- Compare a CRT monitor and an LED monitor side by side if you still have an old one in the store room.
- Plug headphones into the light green port and a microphone into the light pink port in front of the class.
- Ask students which operating system is pre-installed on the lab computers, and whether it is licensed.

MARKING SCHEME

Unit 1 Test

Total 40 marks

1 Choose the correct answer

8 marks (1 each)

1. (c) a system analyst	**2.** (d) Tablet
3. (b) Flash drive	**4.** (c) gigahertz
5. (d) 4	**6.** (b) HDMI
7. (c) light pink	**8.** (b) A and C only

2 Fill in the blanks

8 marks (1 each)

1. peripheral	**2.** brain	**3.** 1000	**4.** 1024
5. inches	**6.** secondary	**7.** extended	**8.** RJ45

3 Look at the pictures

8 marks (1 + 1 each)

1. HDMI port - sends sound and video (to a monitor, TV or projector)	**2. CRT monitor** - uses more electricity, is heavy and takes more space
3. Dual core processor - has two processors (CPUs) in it	**4. Network port (RJ45)** - connects the computer to a network

1 mark for the name and 1 mark for what it does or a correct fact.

4 Answer these

16 marks

1. Name the two groups of computers according to their nature and use. Give two examples of each. (3 marks)

Non-portable computers - any two of server, workstation, desktop, all-in-one; computers for mobile use - any two of laptop, notebook, tablet, smart phone. 1 mark for the two names, 1 mark for each pair of examples.

2. Why is it advisable to select a processor with a greater speed? (2 marks)

It processes more data in a unit time (1), so the performance of the computer increases and software runs faster (1).

TEACHING TIP

- Read the warranty card of a school computer or printer with the class. Find the warranty period.

3. Name three technical specifications and three non-technical specifications of a computer. (3 marks)

Technical: processor speed, hard disk capacity, memory capacity, screen size (also processor type, monitor type, VGA, sound card). Non-technical: price, warranty, company reputation, after sales service. 1 mark for each correct pair (one technical and one non-technical).

4. Explain the three kinds of warranty. (3 marks)

Manufacturer warranty - given by the manufacturer; the device is repaired or replaced. Extended warranty - given in addition to the standard warranty, at an additional cost. On-site warranty - technicians visit the place of work to repair the product. 1 mark each.

5. A projector is connected with an HDMI cable, and another one with a VGA cable. What is the difference? (2 marks)

HDMI sends both sound and video signals (1). VGA sends only images, so a separate cable is needed for sound (1).

6. Name the three after-sales services given by computer sales centres. (3 marks)

Technical advice; technical assistance; telephone, e-mail and website information (contact information) of the vendor. 1 mark each.

WHAT STUDENTS USUALLY GET WRONG

- They call the touch screen only an output device. It is an input **and** an output device.
- They write 1 GB = 1000 MB. For memory it is **1024 MB = 1 GB**; for speed it is 1000 MHz = 1 GHz.
- They compare 3.2 GHz and 3000 MHz without changing the units first.
- They measure a monitor across its width. The size is the **diagonal** of the screen, in inches.
- They think the main memory keeps data permanently. It keeps data and instructions **temporarily**.
- They say a VGA cable carries sound. Only **HDMI** sends sound and video together.
- They choose a computer by its price only. The textbook says that is **not advisable**.

HOMEWORK AND EXTRA PRACTICE

- Cut out a computer advertisement and mark each specification T (technical) or N (non-technical).
- Write the specification of the computer you would like to have, and the reason for each choice.
- Ask at home: which devices have a warranty card? Write the warranty period of each.
- Fast learners: compare the price of the same laptop in three newspapers or websites and report.

TEXTBOOK NOTES

- Page 13 says the main memory is measured "in units of bits (Mega Bite (MB) or Giga Bite (GB))". The units are **bytes**: megabytes and gigabytes. The workbook says bytes.
- Page 13 prints "DV1"; it is **DVI** (Digital Visual Interface). Page 15 prints "trail version"; it is a **trial** version.
- The table of page 20 lists "Generation" under main memory. The workbook copies it but does not test it.
- The summary (page 21) calls pre-installed software a non-technical factor; section 1.5.7 teaches it as a specification. The workbook does not ask which group it belongs to.
- Table 1.4 puts Core i3, i5, i7 and i9 under quad core. The workbook follows the textbook.
- The textbook sends students to a separate workbook for Activities 1.1 - 1.11. Here they are Practicals 1 - 10 (Activities 1.6 and 1.7 are one practical, as the textbook names them together).

TEACHING TIPS

- Ask students to bring warranty cards or receipts of devices at home (phones, TVs, radios).
- Role play a computer shop: one group sells, one group buys with a specification in hand.
- Show the Bluetooth and Wi-Fi icons on a phone or a laptop. Both give cable free connections.
- For the specification tree, ask students to say the example value before the name of the specification.
- Mark the unit test together the next day. Students learn most from the questions they got wrong.
- Students who finish early can write a specification for a teacher, a shop owner or a gamer.
- Use local prices in rupees from newspapers. Prices change, so let students find this week's prices.
- Remind students: never buy a computer by its price only. Ask them to explain why in their own words.
- Weaker readers: read the exercise sentences aloud. The words are all in the word list.
- Use the quick oral quiz at the start of each lesson to revise the last lesson.
- Practicals 1, 6 and 7 can be done as homework with the help of parents and newspapers.
- This unit prepares students for Grade 10 Unit 2 (the computer system). Keep their work for revision.
- Ask students to explain each analogy of the textbook: brain, blender, exercise book, clock and bag rack.
- Ask students to try one of the five ways to learn more about computers (page 5) at home and report.